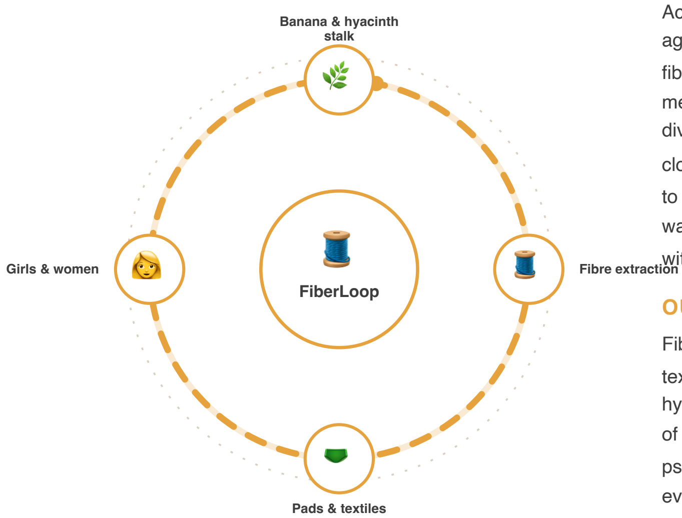
AGRI VALUE-CHAIN

🧵 FiberLoop

Natural-fibre care

Waste-plant fibre from banana and water hyacinth into industry and biodegradable pads



🌿 Banana & hyacinth stalk → 🧵 Fibre extraction → 🩲 Pads & textiles → 👩 Girls & women

THE SITUATION IN BANGLADESH

Across Bangladesh, banana pseudostems, water hyacinth and pineapple leaves are treated as agricultural and aquatic waste, even though banana pseudostem is a lignocellulosic, textile-grade fibre [1] and water hyacinth fibre can be processed into safe absorbent material [2]. At the same time menstrual-hygiene need is acute: a cross-sectional study of 586 adolescent girls in the Rajshahi division found only 37.7% continuously used sanitary pads, with cloth users commonly reusing cloths [4], and poor menstrual hygiene management among rural school-going girls has been linked to lower school performance and absenteeism [7]. The raw material is abundant and currently wasted, the global natural-fibre market it could feed was worth roughly USD 62.87 billion in 2025 with Asia Pacific holding about 51% of it [5], yet little of this value is captured locally.

OUR PITCH

FiberLoop decorticates waste-plant fibre into two revenue streams: industrial natural fibre for textiles, paper/board, packaging and composites [16], and biodegradable sanitary pads for the local hygiene gap. Mechanical decortication is the wedge — a single machine yields roughly 20–30 kg of banana fibre per day versus only ~4 kg by hand, a 5–7× productivity gain that turns discarded pseudostems into saleable fibre and rural income [8]. The pad application is grounded in real evidence: a 2026 peer-reviewed study made water-hyacinth pads that absorbed 10 mL in ~3 seconds at a skin-neutral pH of 6.87, stayed microbially safe, and biodegraded ~95% in 60 days [2], while Saathi has already commercialized compostable banana-fibre pads that decompose in 3–6 months [3]. We are honest that the industrial fibre and decortication economics are proven [18], whereas FiberLoop's own at-scale pad manufacturing in Bangladesh remains aspirational and must still clear local safety validation and unit-cost targets before claiming the lab and Saathi precedents as our own [23].

SOURCES (8)

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